Parallel Computing Project Report: *Wave Function Collapse*

Filippo Battisti, Simone Freschet, Francesco Bittasi

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1 Introduction

Introduce the problem. Cite the canonical reference / textbook. State the input size n and the relevant cost model.

Outline the sections that follow.

Cite [1]

2 Sequential algorithm

2.1 Problem definition

Formal definition: inputs, outputs, constraints, notation used throughout the report.

2.2 Algorithm and complexity

Describe the standard sequential algorithm you parallelize. State its time and space complexity and, where relevant, its data-dependency structure that is what your parallel design has to respect.

Algorithm 1 Sequential your super cool algorithm

1: *// Implementation details*

3 Parallel design

3.1 Dependency analysis

Identify independent work units and the partial order between them. Distinguish "embarrassingly parallel" phases from synchronization points.

3.2 Communication pattern

List each collective / point-to-point step, who talks to whom, and the message volume per step.

4 Topology-aware implementation

4.1 Topologies considered

[State the topologies you implement and why. Examples (non-exhaustive): linear array, ring, 2D mesh, 2D torus.]

4.2 Embedding the communication pattern

[Topology 1]. [How the algorithm's communication maps onto this topology.]

[Topology 2]. [Same as above for the next topology. Highlight the qualitative difference.]

5 MPI implementation notes

5.1 Language and build

[State the language (C, C++, Rust, Java, Python via mpi4py, …) Why this language?]

5.2 Topology realisation in MPI

Discuss how each topology is realised and the typical MPI primitives used and justify why.

6 Experimental setup

Hardware. Machine(s) used: CPU model, core count, memory, interconnect (if multi-node).

[If single-machine, state that all "processes" map to cores of the same node and that the topology is therefore logical, not physical.]

Inputs. [How inputs were generated (distribution, seed), input sizes tested, and any preprocessing.]

Methodology. [Number of repetitions per configuration, what is averaged or reported, how time is measured, and how compute time is separated from communication time.]

7 Results

[Brief description and table and/or plot for each section]

7.1 Execution time

7.2 Speedup and efficiency

7.3 Comparison across topologies

7.4 Computation vs. communication

7.5 Scalability

8 Discussion

Interpret the results, do not just describe them.

8.1 Limitations

[Honest assessment: what the experiments do not show, what would need a multi-node cluster, what artifacts may arise from running all topologies on a single machine, etc.]

8.2 Possible improvements

[Concrete next steps]

9 Conclusions

[5-10 sentences. The headline result and the main lesson learned.]

References

[1] Thomas H Cormen et al. *Introduction to algorithms*. MIT press, 2022.